

Distributed System (CT 703) – Past Paper Frequency Analysis

Chapter 1 – Introduction

(4 hrs / 8 marks)

Q1 is always from this chapter across all 19 sets. The definition is asked every time; the second part rotates through the sub-topics below.

Topic	Frequency
Definition of DS (present in every Q1; sub-aspects below)	19
DS vs centralized system — advantages, disadvantages, comparison	7
Goals / design goals / importance of DS	6
Design challenges / requirements of DS	6
Models / architecture of DS (interaction model, hardware/software distribution)	5
Transparency — types, layers, justification	4
Fundamental model of DS	2
Heterogeneity in DS	2

⚠ Near-exact repeats:

- "What are the major goals of DS and challenges during design?" → 2076A, 2069C (same question)
- "Define DS. What are advantages and disadvantages?" → 2070C, 2073S (same core)
- "Why DS preferred over centralized?" → 2080B, 2075C, 2072C, 2070C (same ask, different framing)
- "DS acts as a single coherent system. Justify with features and challenges." → 2071C; near-same in 2080Ba

Chapter 2 – Distributed Objects and File System

(7 hrs / 16 marks)

This chapter produces two questions in almost every paper, consistent with its 16-mark allocation.

Distributed Objects side:

Topic	Frequency
RMI — static, dynamic, architecture, operation	14
RPC — functional steps, communication semantics, mechanism	11
RPC vs RMI — comparison	10
IDL role in distributed system	4
Events and notifications in distributed object communication	2

File System side:

Topic	Frequency
DFS definition + file service architecture (draw and explain)	19
SUNNFS / modern DFS — architecture and operation	9
DNS working mechanism (recursive/iterative queries, with example)	7
Stateful vs stateless DFS	6
Role of DFS in resource sharing	3

 Exact/near-exact repeats:

- "What is DNS? Explain DNS working mechanism with suitable example." → 2075A-Q3 ≈ 2074A-Q3 (near-identical)
- "Compare stateful and stateless DFS. Explain working of one modern DFS with architecture." → 2081B-Q3, 2080Ba-Q3, 2075C-Q3 (same template, three sets)
- "Differentiate between RPC and RMI." → core ask in 2078B, 2075C, 2074A, 2072C, 2070C, 2070A (6 sets)

- "Explain file service architecture for DFS. Define queries and operations of DNS." → 2080B-Q3
≈ 2073S-Q2 (same structure)

Chapter 3 – Operating System Support

(3 hrs / grouped with Ch4+10 for 16 marks combined)

Ch3 rarely appears as a standalone main question. Most appearances are as part (a/b) of compound questions or as short notes.

Topic	Frequency
Monolithic vs microkernel OS architecture	6
Network OS vs Distributed OS — key differences, NOS preferred over DOS	5
Processes vs threads in DS — compare, why threads important	5
Distributed OS (DOS) — characteristics and definition	3

Chapter 4 – Distributed Heterogeneous Applications and CORBA

(3 hrs / grouped with Ch3+10)

Topic	Frequency
CORBA architecture + services + ORB + Object Adapter (diagram)	17
Role of middleware in DS	4
Dynamic vs static invocation approaches in CORBA	3
Heterogeneous vs homogeneous distributed applications	2

⚠ CORBA appears in every single paper — either as a main question or a short note. The one guaranteed topic from this chapter in any given exam. Know the full component diagram: IDL → client stub → ORB → object skeleton → Object Adapter → CORBA services.

Chapter 5 – Time and State in Distributed Systems

(5 hrs / 8 marks)

Topic	Frequency
Lamport's logical clock — principles, implementation rules, limitations, algorithm with example	13
Physical vs logical clock — differentiation	8
Physical clock synchronization — NTP and Berkeley Algorithm	6
Vector clock — implementation rules, causality of events, comparison with Lamport	6
Global state recording / Chandy-Lamport snapshot algorithm	5
CUT types (consistent vs inconsistent) in distributed computation	3
Cristian's Algorithm	3
Distributed debugging	3
Causal ordering of messages	2

⚠ Exact/near-exact repeats:

- "What is the issue with Lamport's clock? Explain alternate (vector clock) algorithm." → 2081B-Q4, 2076C-Q4 (same concept)
- "Write implementation rules of Lamport clock. State limitations." → 2078B-Q5 ≈ 2076A-Q5
- "Difference between physical and logical clock. Discuss principles of Lamport's clock with algorithm." → 2081Ba-Q4 = 2074A-Q5 (near-identical)
- "Vector timestamp mechanism for causality of events. Justify with implementation rules and examples." → 2080B-Q5 ≈ 2080Ba-Q4 (consecutive sets, same question)

Chapter 6 – Coordination and Agreement

(4 hrs / 8 marks)

Topic	Frequency
Mutual exclusion — requirements, token vs non-token comparison	13

Topic	Frequency
Election algorithms — Bully, Ring-based, Central Coordinator	10
Ricart-Agrawala algorithm (token and non-token variants)	8
Token Ring algorithm for mutual exclusion (with algorithmic steps)	5
Consensus in DS	5
Multicast / reliable group communication	3

⚠ Exact/near-exact repeats:

- "Compare token-based and non-token-based mutual exclusion." → 2080Ba, 2078B, 2075C (same core, three sets)
- "Explain any one election algorithm with example." → 2070C, 2070A, 2069C (same ask, three sets)
- Ricart-Agrawala as the non-token based algorithm to explain → 2075A, 2074A, 2072C, 2071S (4 sets, same algorithm by name)
- "What are the fundamental requirements of mutual exclusion? Why is election applicable in DS?" → 2081B-Q5, 2072C-Q6 (same structure)

Chapter 7 – Replication

(4 hrs / 8 marks)

Topic	Frequency
Active vs passive (primary-backup) replication — compare and implement	16
Reasons for replication / replication as scaling technique	9
Fault tolerant services through replication	7
High availability services	4
Consistency models in DS	2
Gossip architecture	1

⚠ Active vs passive replication is asked in **16 out of 19 papers** — the single most consistently tested topic in the entire subject. Memorize both models with architecture diagrams.

⚠ Exact/near-exact repeats:

- "What are the reasons for replication? Explain active replication model with its advantages and disadvantages." → 2072C-Q7 = 2075C-Q7 (exact same question, two different sets)
- "Differentiate between passive and active replication. Discuss a technique that makes the DS highly available." → 2075A-Q8 ≈ 2069C-Q7 (near-identical)
- "How does passive replication support fault tolerance? How does it differ from active replication?" → 2079B-Q6 ≈ 2078B-Q7 (same ask, consecutive years)

Chapter 8 – Transaction and Concurrency Control

(6 hrs / 8 marks)

Topic	Frequency
Concurrency control methods — locks, 2PL, strict 2PL, optimistic, timestamp ordering	12
Distributed deadlock — definition, avoidance approaches, phantom deadlock	11
Flat vs nested vs distributed transaction — definitions and comparison	10
Two-phase commit protocol (2PC)	8
Three-phase commit protocol (3PC) — with state diagram	7
Optimistic concurrency control	5
Timestamp ordering for concurrency control	4
Cascading aborts — causes and solutions	3
Two-phase locking (2PL) and strict 2PL	3

⚠ Exact/near-exact repeats:

- "Compare nested transaction with distributed transaction. Explain two-phase commit protocol for handling distributed transactions." → 2081B-Q8 = 2075C-Q8 (exact same wording)

- "How do cascading aborts occur and how are they solved? Explain 3PC with state diagram." → 2074A-Q8 = 2072K-Q8 (near-identical)
- "What are the flat and nested transactions? Explain how problems of 2PC are solved by 3PC." → 2078B-Q9 ≈ 2079B-Q8 (same concept, consecutive years)
- "What are solutions to avoid deadlock in DS? Explain three-phase commit protocol." → 2080B-Q8 = 2076C-Q8 (same structure)

Chapter 9 – Fault Tolerance

(4 hrs / 8 marks)

Topic	Frequency
Fault, error, failure — definitions, types, detection and recovery	8
Forward/backward recovery — checkpointing (coordinated vs independent)	8
Byzantine fault + agreement in faulty systems (Byzantine Generals problem)	7
Process resilience	5
K-fault tolerant systems	3
Distributed commit (in recovery context)	3
Reliable client-server communication	2
Triple modular redundancy	1

⚠ Exact/near-exact repeats:

- "Define fault, error and failure. How do you detect arbitrary faults and recover?" → 2081B-Q9 ≈ 2069C-Q10 (same template)
- "Define faults, failures and errors." → standard opening of almost every Ch9 main question
- "Compare independent checkpointing with coordinated checkpointing." → 2074A-Q8 (context) ≈ 2071S-Q8 (near-identical structure)
- "Explain forward and backward recovery. How to implement coordinated checkpointing?" → 2076C-Q7 ≈ 2078B-Q8 (same concept)

Chapter 10 – Case Studies

(5 hrs / grouped with Ch3+4)

Ch10 topics appear exclusively as short notes. No standalone main question on any case study (except JINI goals partially in 2071C-Q5).

Topic	Frequency
MACH	7
JINI	4
CORBA as case study (dynamic/static invocation depth)	3
TIB/Rendezvous	2

Step 3 – Diagram Questions

Only one genuine diagram question exists across all 19 sets:

Type 1 – CUT Determination (Distributed Global State)

Frequency: 1 (2073 Shrawan Q5 only)

Pattern:

Given: Spacetime diagram with processes P0, P1, P2 and
cuts C1, C2, C3, C4 drawn across the processes

Find: Classify each cut as consistent or inconsistent
Identify strongly consistent CUT if any

Sub-types:

- Consistent CUT: no message crosses the cut from right to left
(i.e., no send is after the cut while receive is before)
- Inconsistent CUT: a message is sent after the cut on the sender side
but received before the cut on the receiver side

 Note: This exact question appeared only once – low exam risk as a standalone diagram question. However, the underlying concept

(CUT types, global state, strongly consistent CUT) is asked as pure theory in 3 other sets (2080Ba-Q9b, 2079B-Q4, 2081B-Q10a).
Know consistent vs inconsistent CUT definition with a drawn example.

Step 4 – Chapter Summary

Ch1 – Introduction

- Chapter type: Theory only
- Most important to practice: All sub-aspects of DS definition — treat goals, challenges, transparency, models, and comparison with centralized as one flexible answer block
- Repeated datasets worth memorizing: "Goals + challenges of DS" combo; transparency types (access, location, replication, failure, migration, concurrency)
- Q1 is always from this chapter — guaranteed

Ch2 – Distributed Objects and File System

- Chapter type: Theory only
- Most important to practice: File service architecture diagram (19/19), RMI static vs dynamic (14/19), DNS with example (7/19)
- Repeated datasets worth memorizing: ⚠ "Stateful/stateless + modern DFS architecture" (3 exact repeats); "Differentiate RPC vs RMI" (6 sets)
- Expect 2 questions from this chapter every exam

Ch3 – OS Support

- Chapter type: Theory only
- Most important to practice: Monolithic vs microkernel comparison; processes vs threads
- No standalone 8-mark question — short note depth is sufficient for all Ch3 topics

Ch4 – CORBA and Middleware

- Chapter type: Theory only
- Most important to practice: CORBA complete architecture with component diagram — ORB, IDL, Object Adapter, dynamic/static invocation, CORBA services

- Repeated datasets worth memorizing: ⚠ Full CORBA architecture — asked in 17/19 papers, never safe to skip

Ch5 — Time and State

- Chapter type: Theory only (one Diagram exception)
- Most important to practice: Lamport's clock rules + limitations (13/19), vector clock rules + causality (6/19), NTP/Berkeley (6/19)
- Repeated datasets worth memorizing: ⚠ "Lamport issue → vector clock solution" is a recurring 6-set pattern; learn both together

Ch6 — Coordination and Agreement

- Chapter type: Theory only
- Most important to practice: Mutual exclusion requirements + Ricart-Agrawala (both variants), at least 2 election algorithms (Bully + Ring-based)
- Repeated datasets worth memorizing: ⚠ Token vs non-token comparison table (8 sets); Ricart-Agrawala by name (6 sets)

Ch7 — Replication

- Chapter type: Theory only
- Most important to practice: Active vs passive replication — absolute priority (16/19)
- Repeated datasets worth memorizing: ⚠ "Reasons for replication + explain active replication with advantages/disadvantages" — appeared identically in 2 sets; appeared structurally in 6+ more
- Single highest-frequency topic in the entire subject

Ch8 — Transaction and Concurrency Control

- Chapter type: Theory only
- Most important to practice: 3PC state diagram, optimistic concurrency control phases, deadlock avoidance approaches, flat vs nested vs distributed transaction comparison
- Repeated datasets worth memorizing: ⚠ "Nested transaction + 2PC" (2 exact repeats); "Cascading aborts + 3PC state diagram" (2 exact repeats)

Ch9 — Fault Tolerance

- Chapter type: Theory only

- Most important to practice: Fault → error → failure chain, Byzantine Generals problem, coordinated vs independent checkpointing
- Repeated datasets worth memorizing: "Define fault/error/failure + fault handling" is the standard Ch9 opening in 8 sets — this definition block is always needed

Ch10 — Case Studies

- Chapter type: Theory only (short notes only)
- Most important to practice: MACH (7 times), JINI (4 times)
- Prepare 4–5 sentence summaries only. Never expect a full 8-mark question on any Ch10 topic.